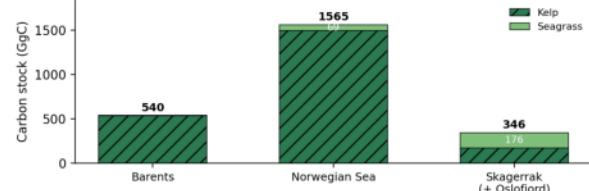


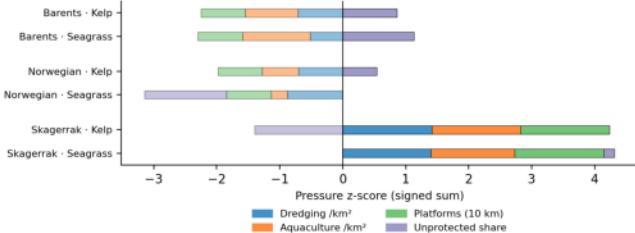
Norway blue carbon — integrated priority synthesis

Stock × Risk × Co-benefit, three canonical regions × two habitats

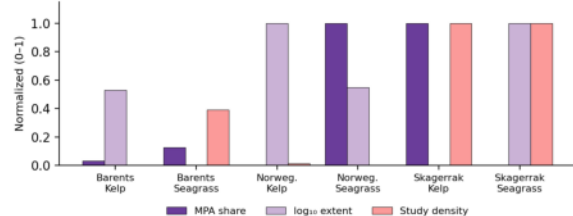
A. Where the carbon is



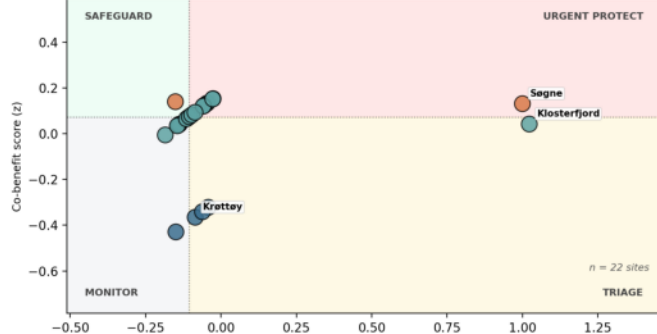
B. Where the pressure is



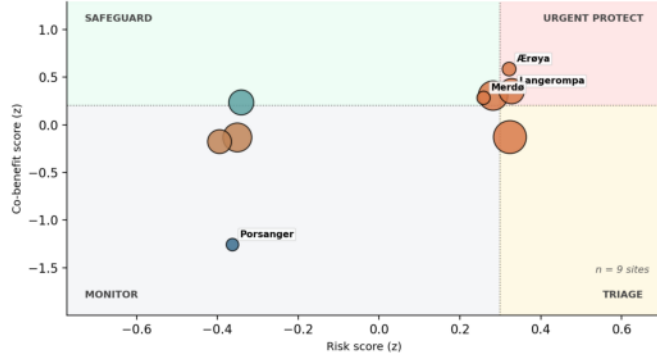
C. Co-benefit fingerprint



D. Priority quadrant — Kelp sites



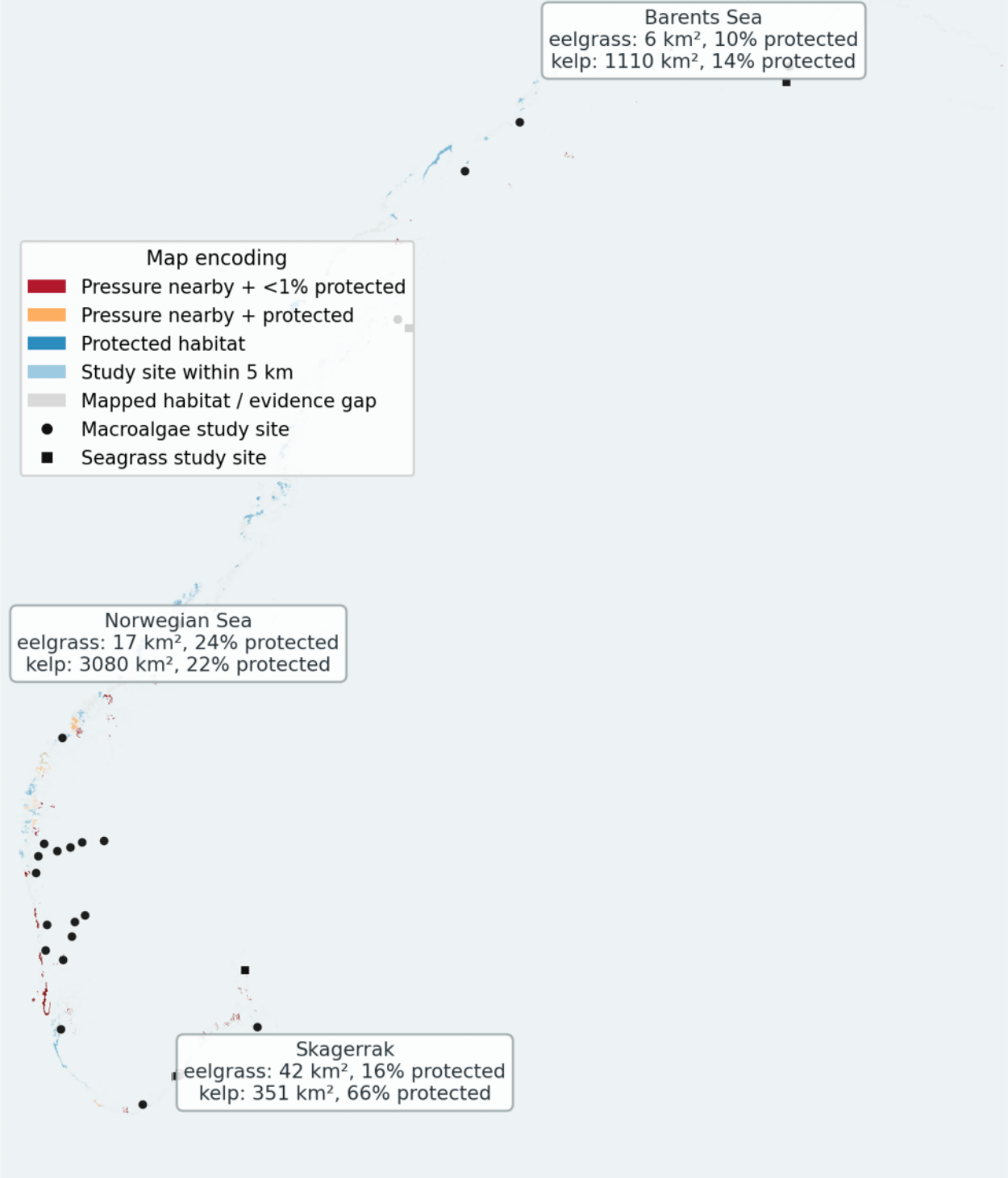
Priority quadrant — Seagrass sites



Panels A-C are regional aggregates (3-region cut: Skagerrak absorbs Oslofjord). Panel D plots individual study sites (n = 31): risk = dredging + akvakultur + platforms within 5 km (haversine), co-benefit = regional MPA share + within-5 km research density + log regional habitat extent, z-scored within ecosystem; small jitter applied so stacked sites are visible. Kelp site stock data unavailable (Ribeiro 2022 pop-genetics) — markers uniform. Seagrass marker size = site carbon stock g C/m² (Gagnon 2024).

Norway Blue-Carbon Co-Location Map

Mapped eelgrass and kelp habitat classified by protection, pressure and evidence coverage



Topline co-location results

Eelgrass

65 km² mapped habitat
17% overlaps protected areas
588 polygons have pressure + <1% protection
93% lack study sites within 5 km

Kelp forests

4,541 km² mapped habitat
23% overlaps protected areas
377 polygons have pressure + <1% protection
98% lack study sites within 5 km

Regional signal

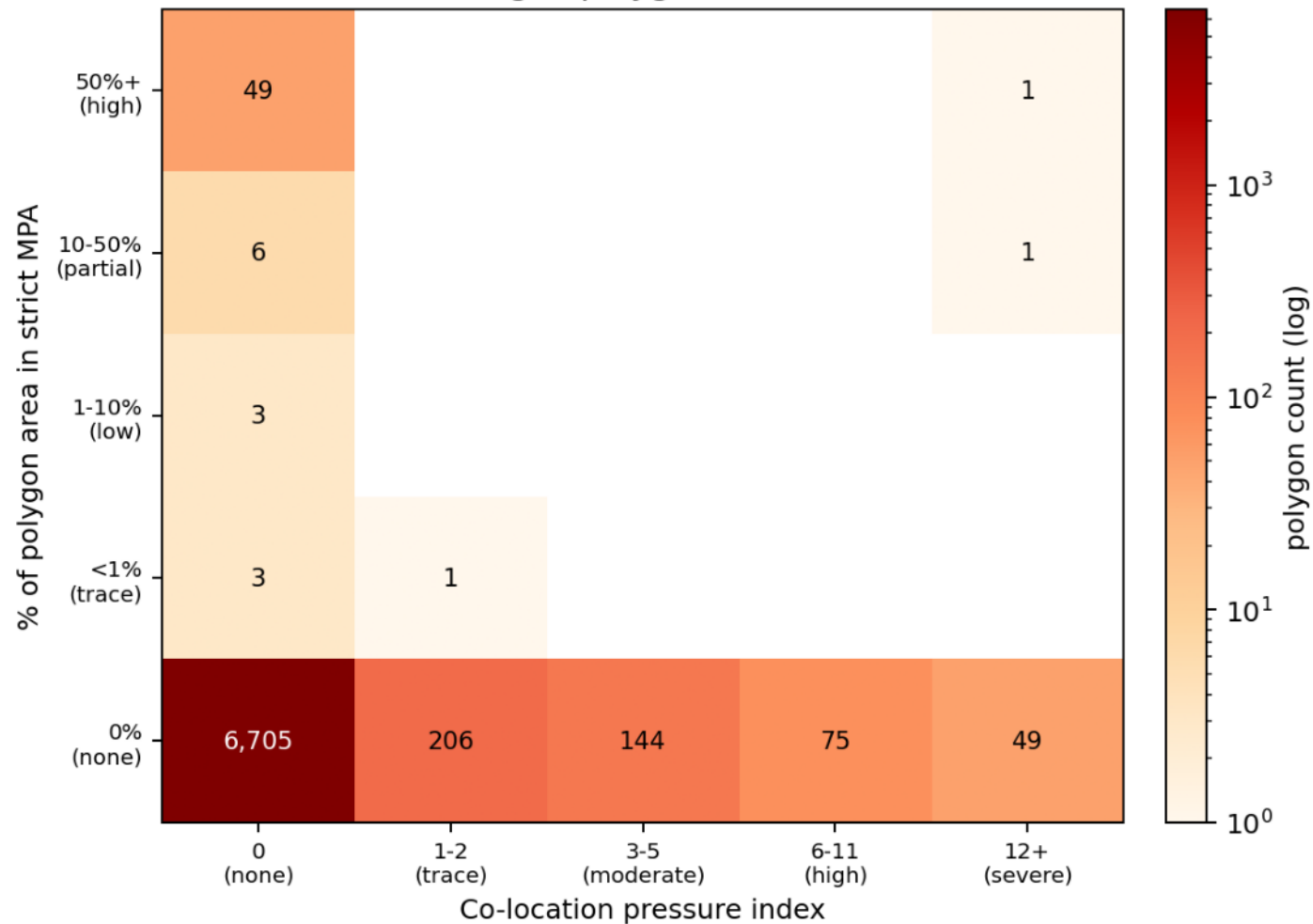
Skagerrak kelp has the highest protection overlap (66%).

Most mapped eelgrass polygons are in Skagerrak (3,310 polygons), but MPA overlap is only 0.1%.

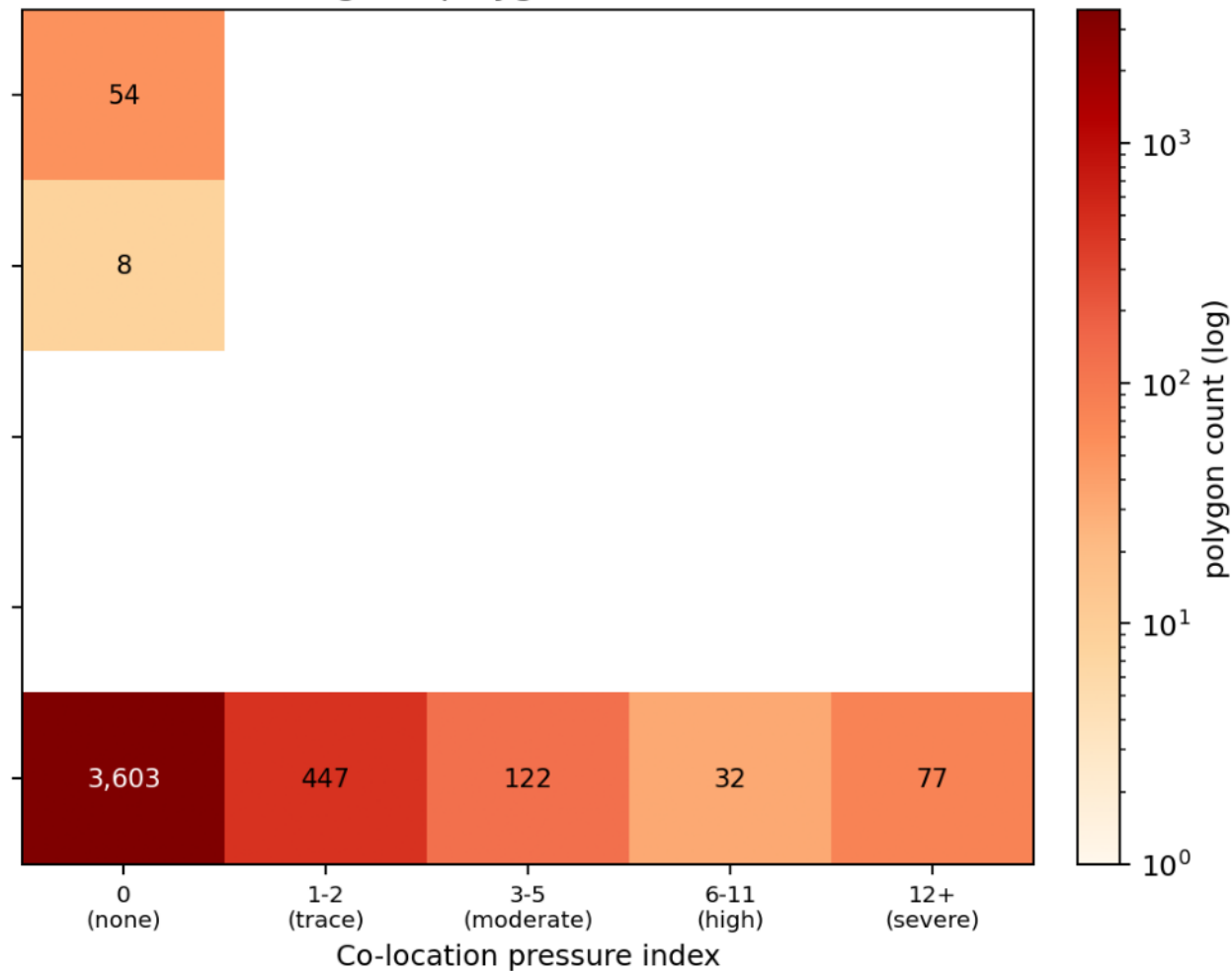
Pressure score is a transparent screening index from dredging, aquaculture, platforms and windfarms. It is not an ecological impact model.

Risk matrix: pressure x strict marine protection (per HB19 polygon)

Macroalgae polygons (n=7,243)

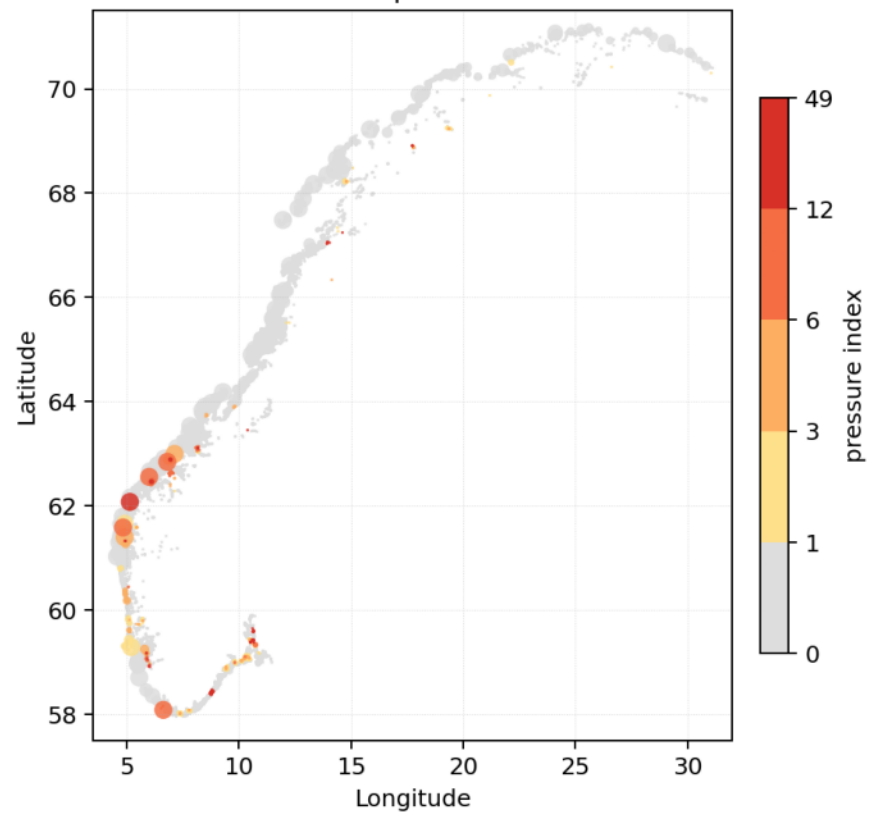


Seagrass polygons (n=4,343)

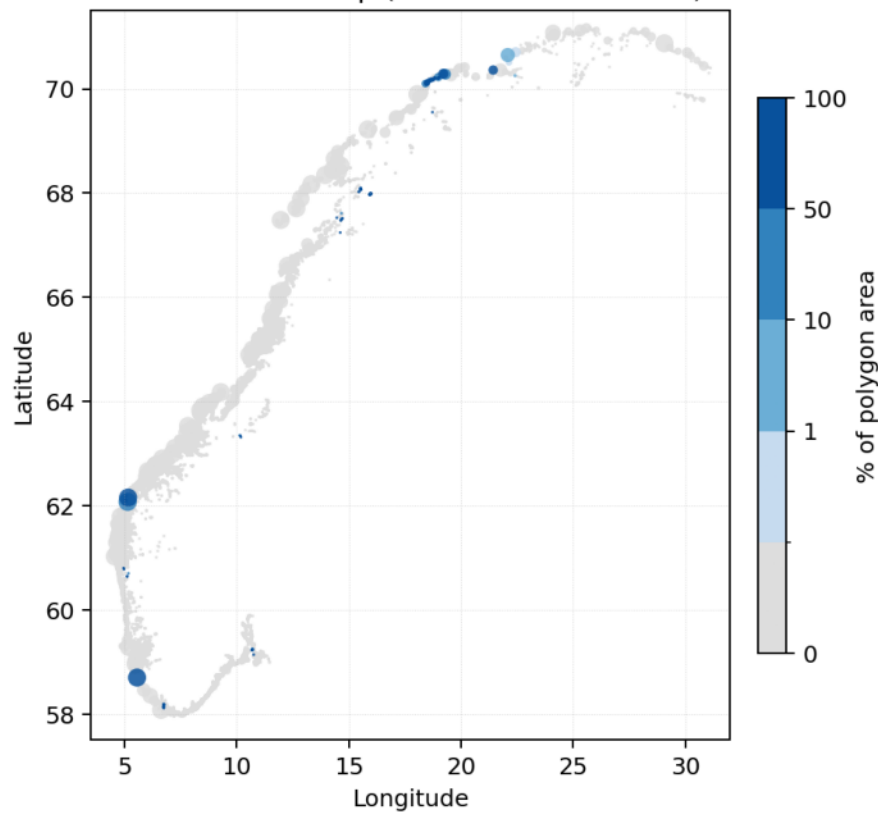


Pressure and protection per HB19 polygon (centroids; size ~ habitat area)

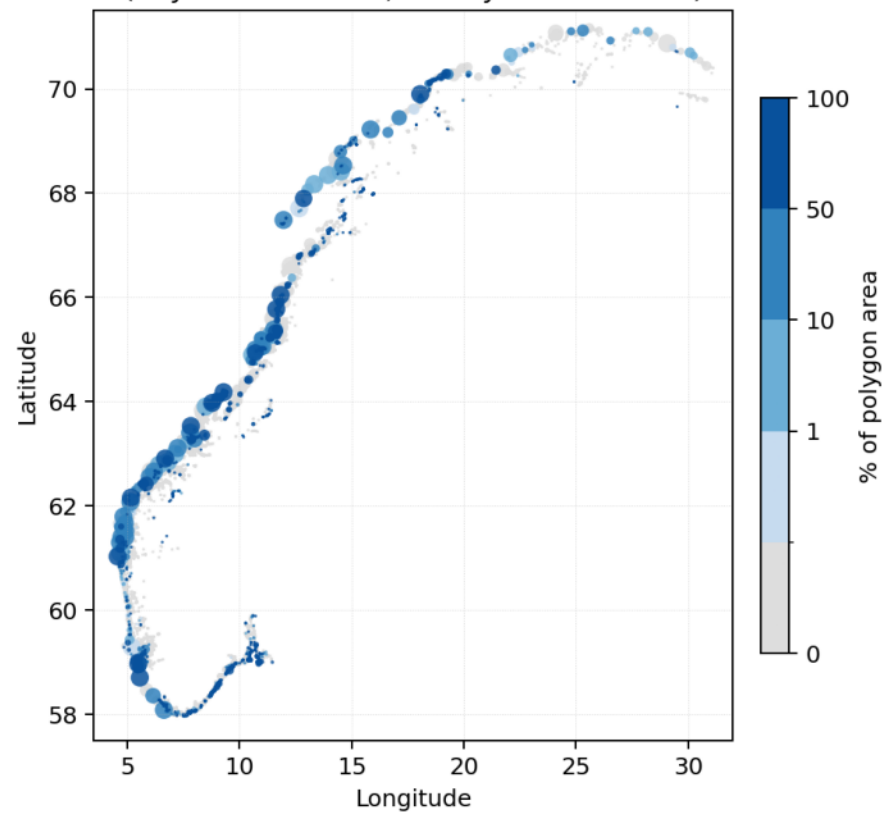
Co-location pressure index



Strict MPA overlap (MarintVerneomraade)

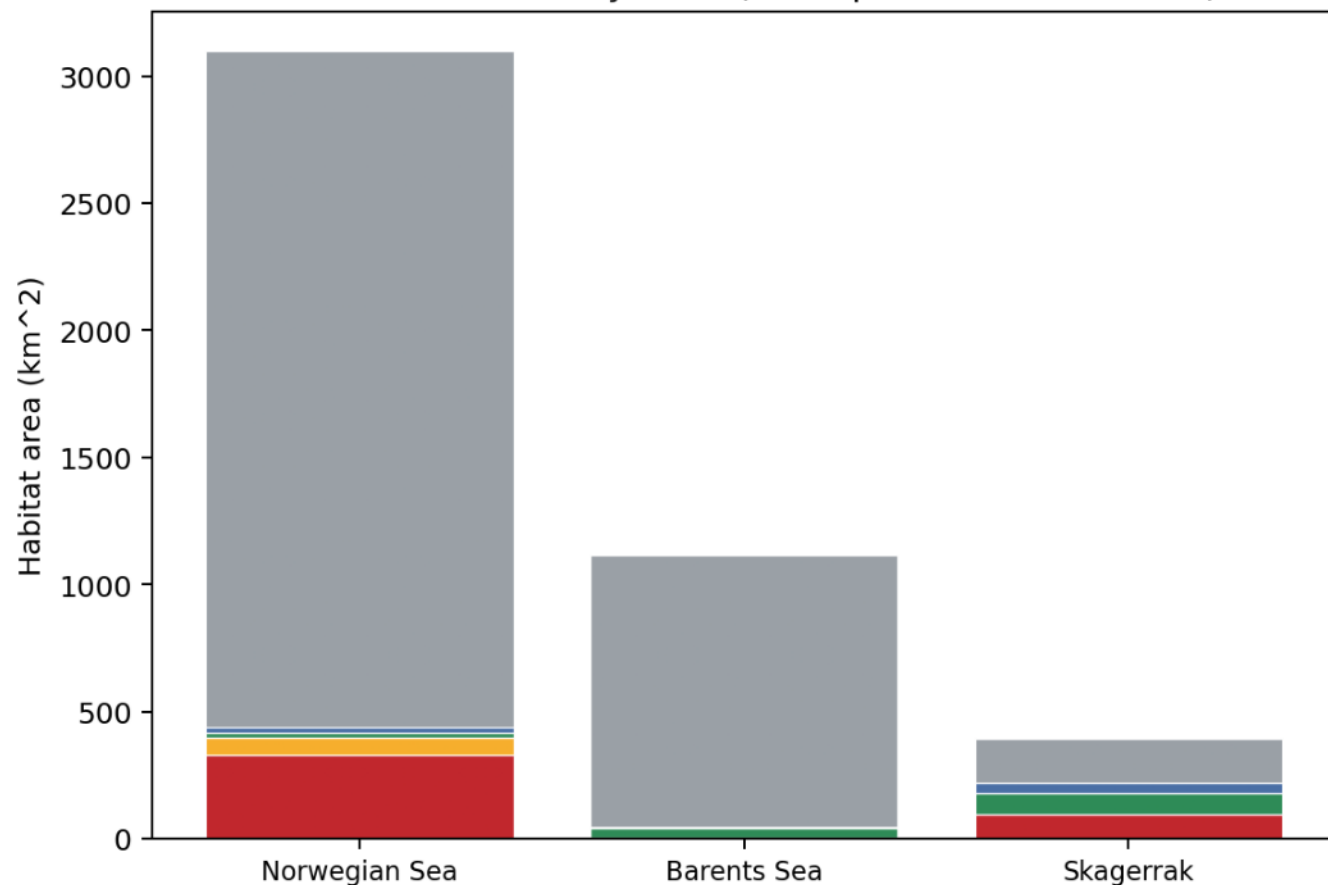


Permissive overlap
(any verneomrade; mostly bird reserves)

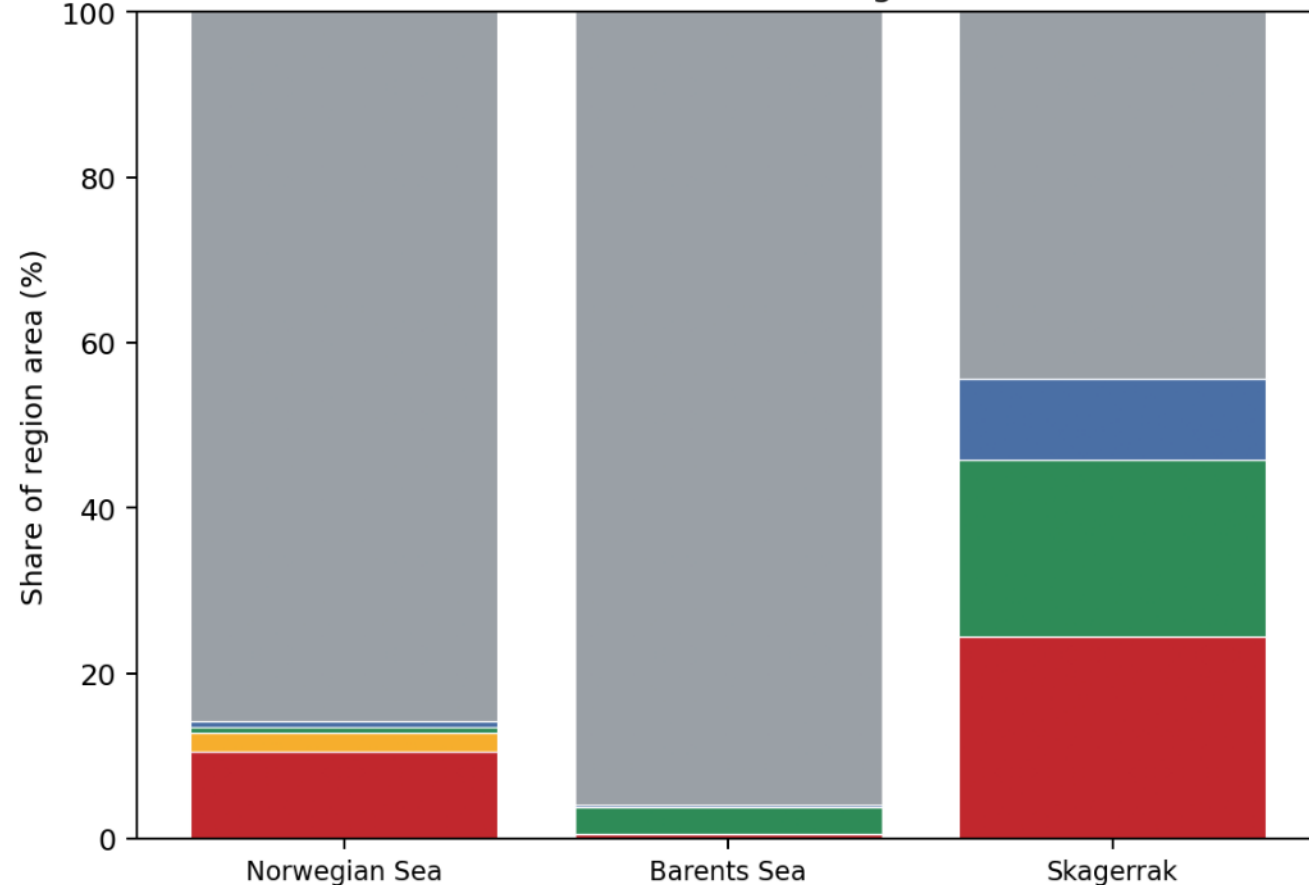


Regional breakdown of co-location classes

Total habitat area by class (strict-protection definition)

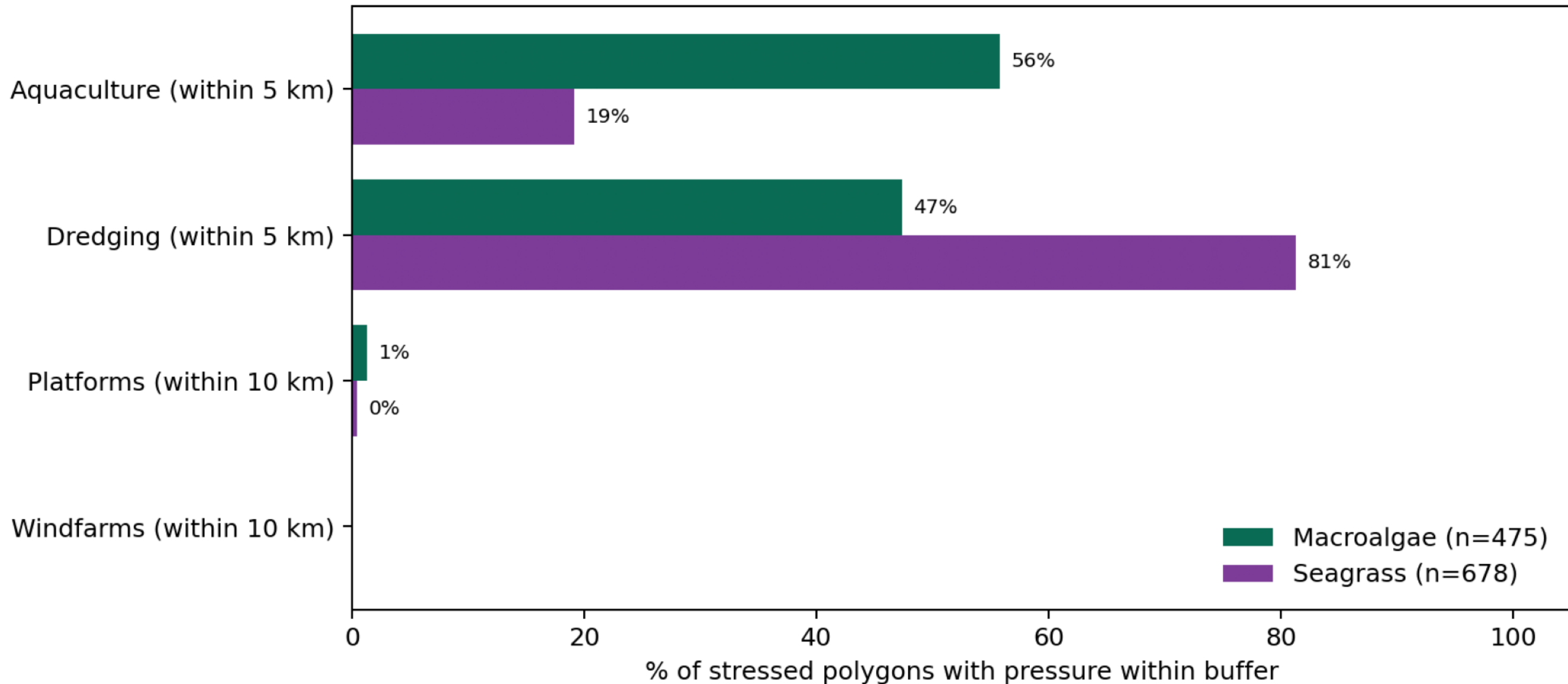


Class share within region



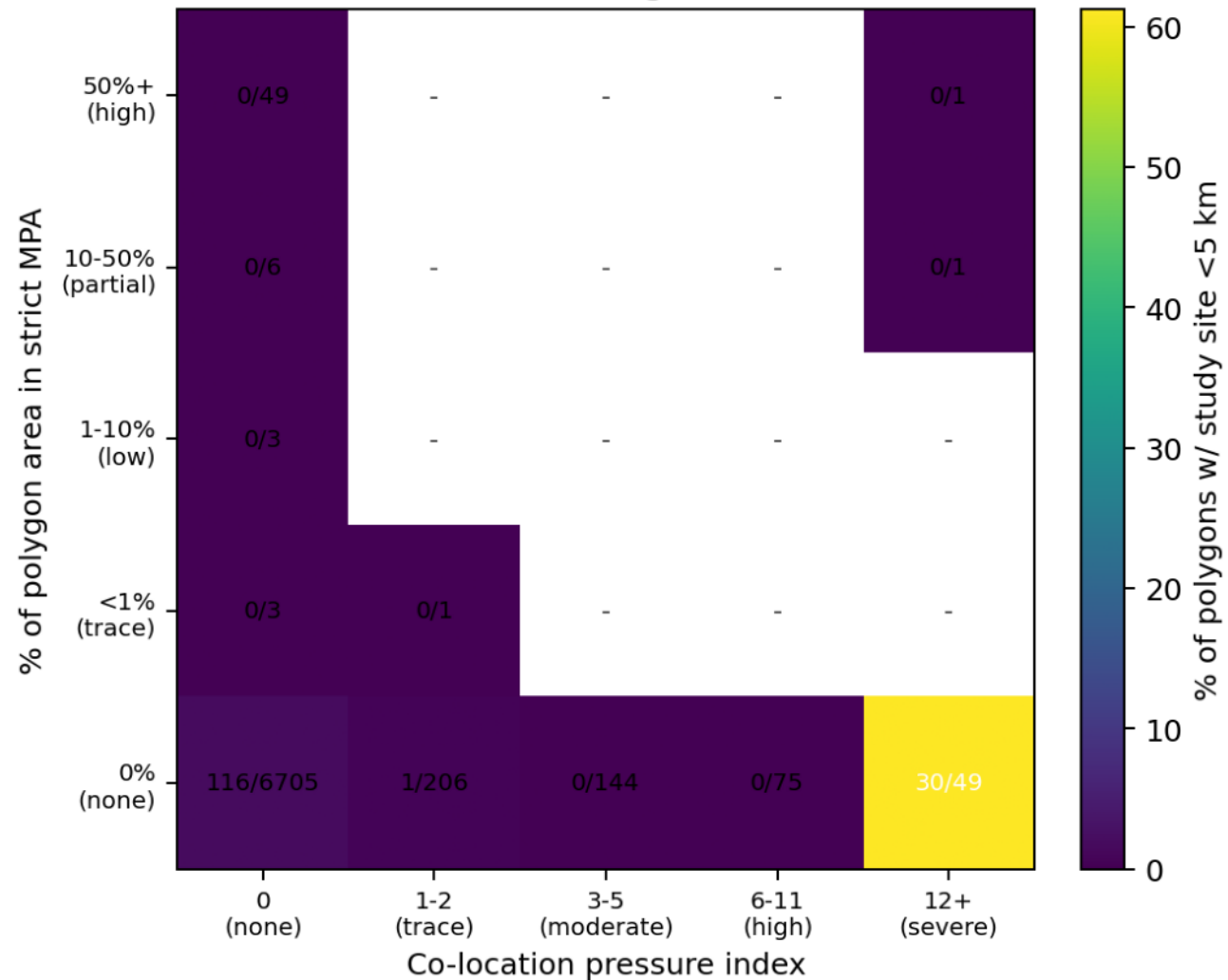
■ Pressure, <1% strict MPA ■ ≥10% strict MPA, no pressure ■ Mapped, no other signal
■ Pressure, ≥10% strict MPA ■ Study site within 5 km

Pressure-type mix among stressed polygons
(pressure > 0 AND <1% in a strict marine MPA)

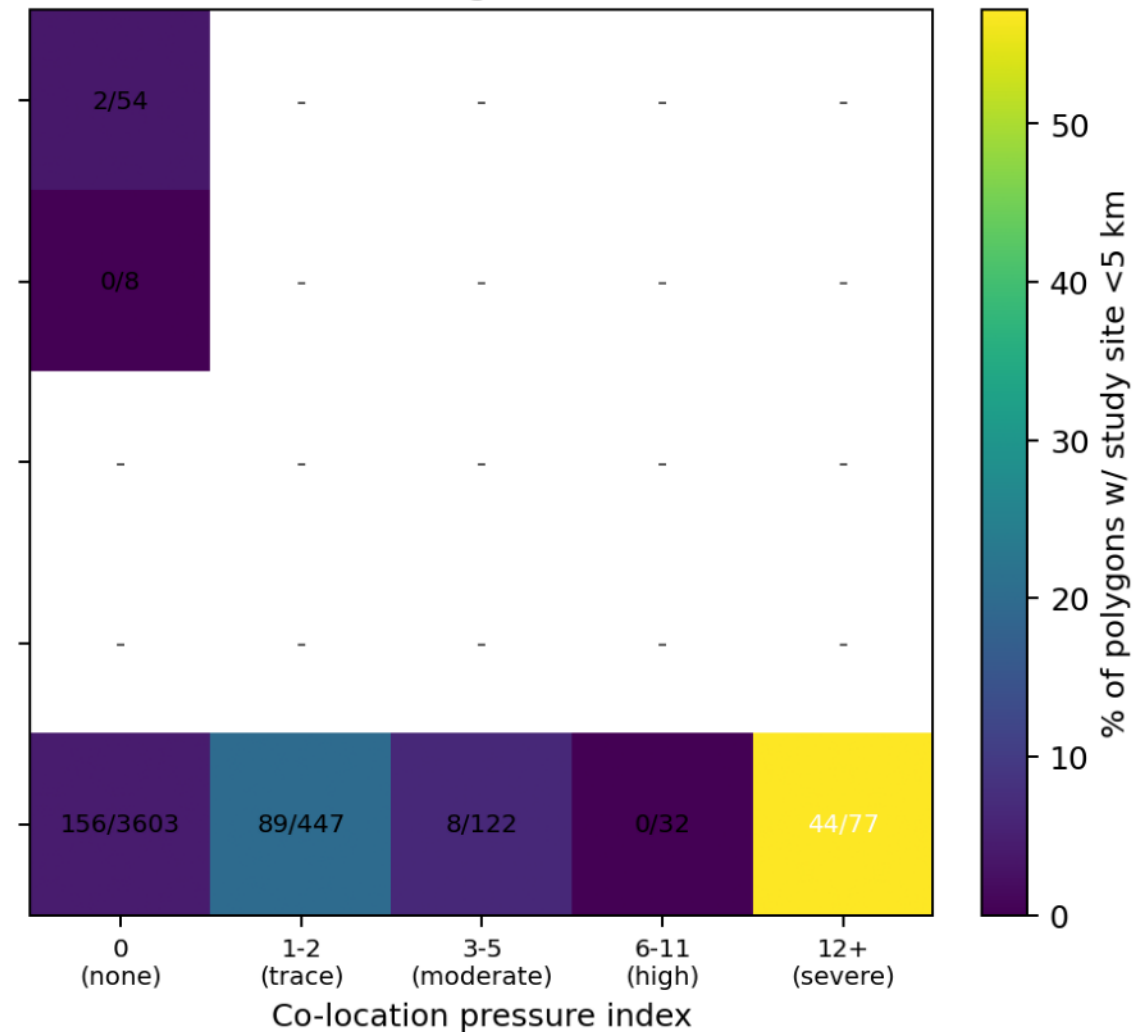


Sampling coverage of the risk matrix
(cells: 'studied / total'; color = study-site coverage %)

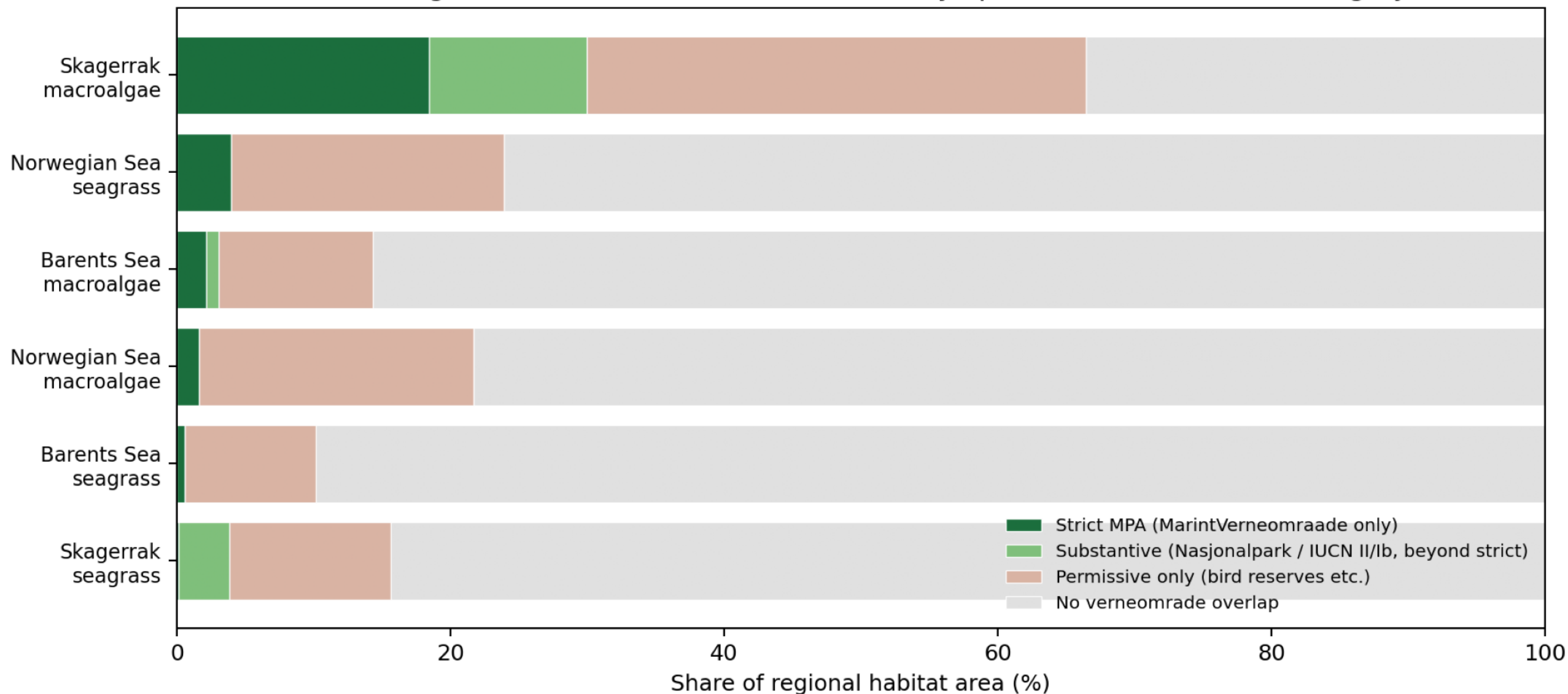
Macroalgae



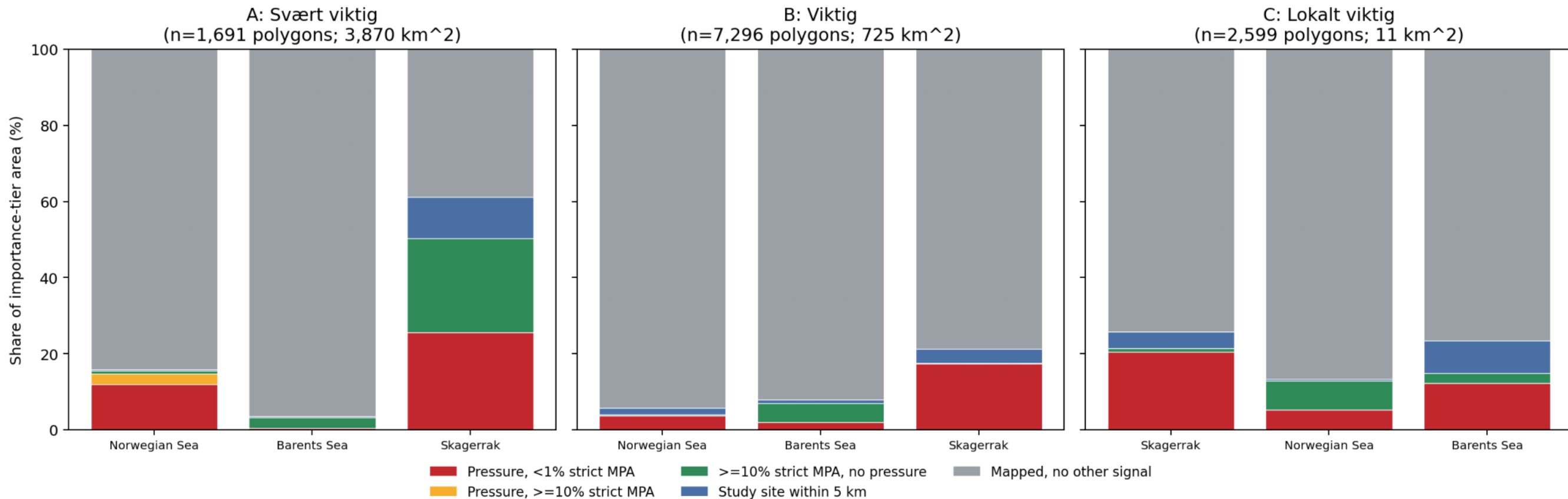
Seagrass



Protection coverage collapses under stricter definitions
(area-weighted; MarintVerneomraade is the only operational marine MPA category)

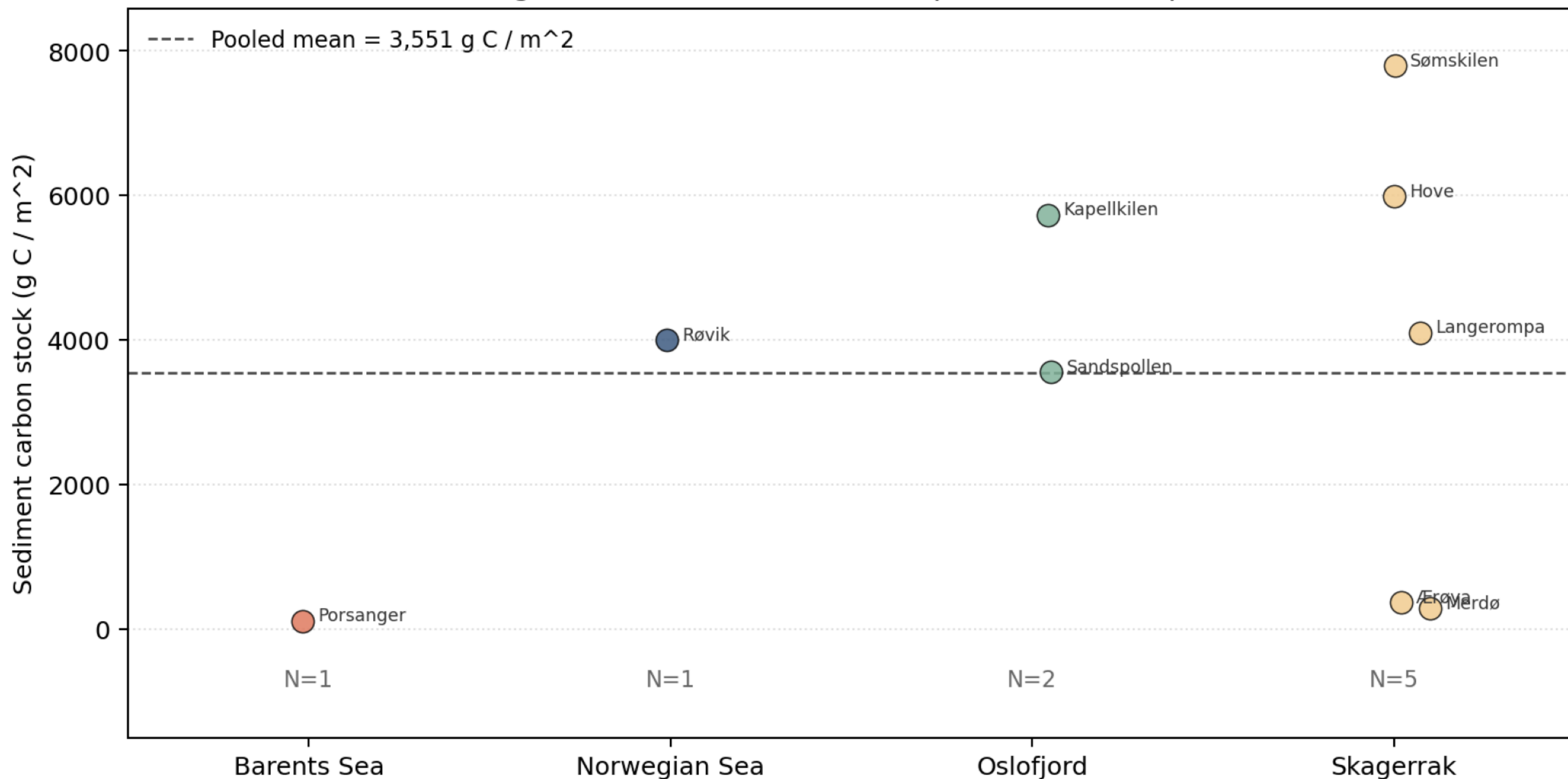


Stress class by HB19 importance tier (area-weighted)
A = Svært viktig (very important) ... C = Lokalt viktig (locally important)



Norwegian seagrass: site-level sediment carbon-stock measurements (N=9 sites)

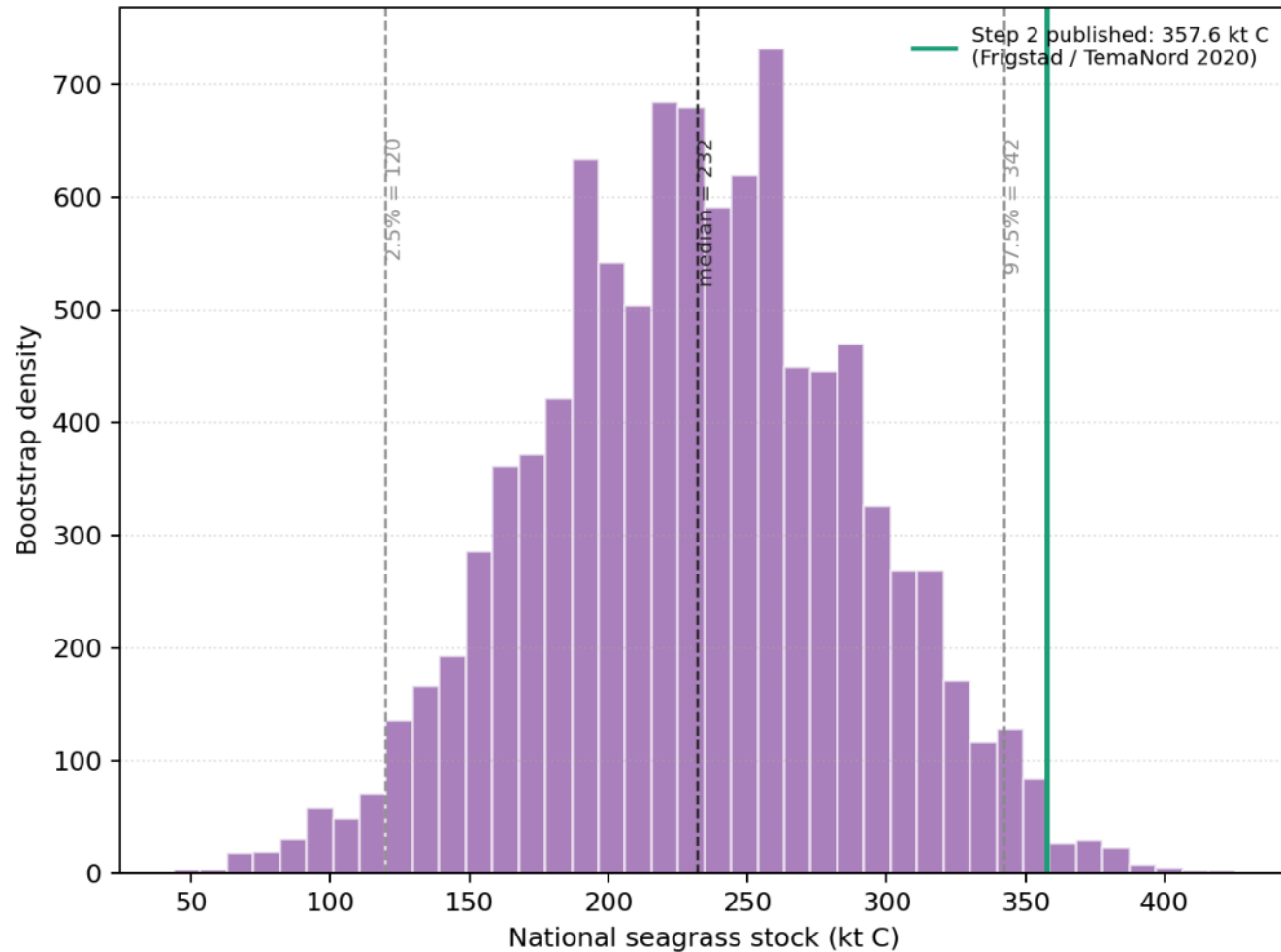
Source: Gagnon 2024, Rohr 2018 (compiled in roadmap workbook)



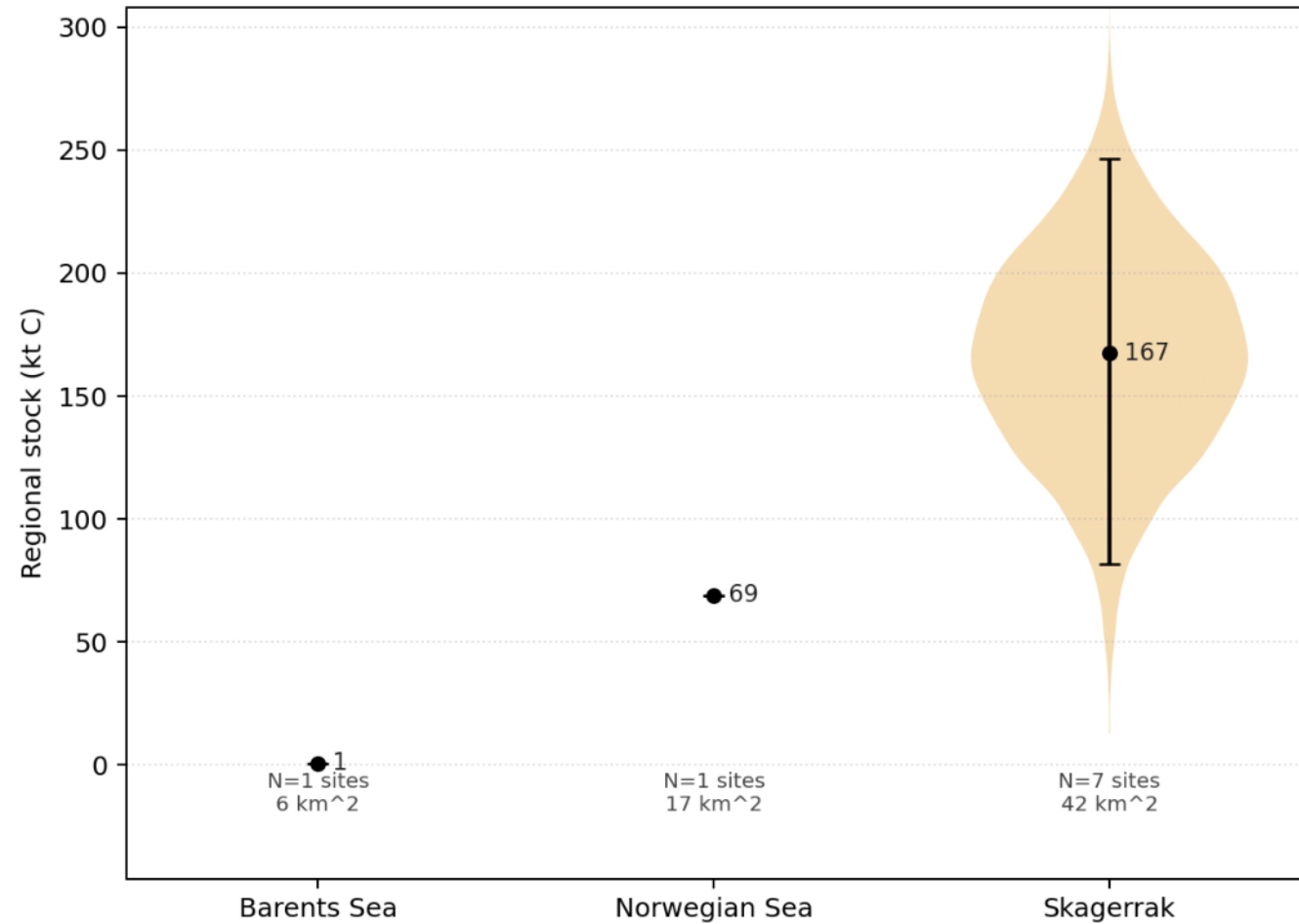
Seagrass national carbon stock - bootstrap uncertainty (95% CI)

left: pooled across all 9 sites; right: per-region area-weighted (Oslofjord measurements pooled with Skagerrak HB19 area; N=1 regions have no within-region variance)

Pooled bootstrap (N=9 sites x HB19 extent 65 km²)

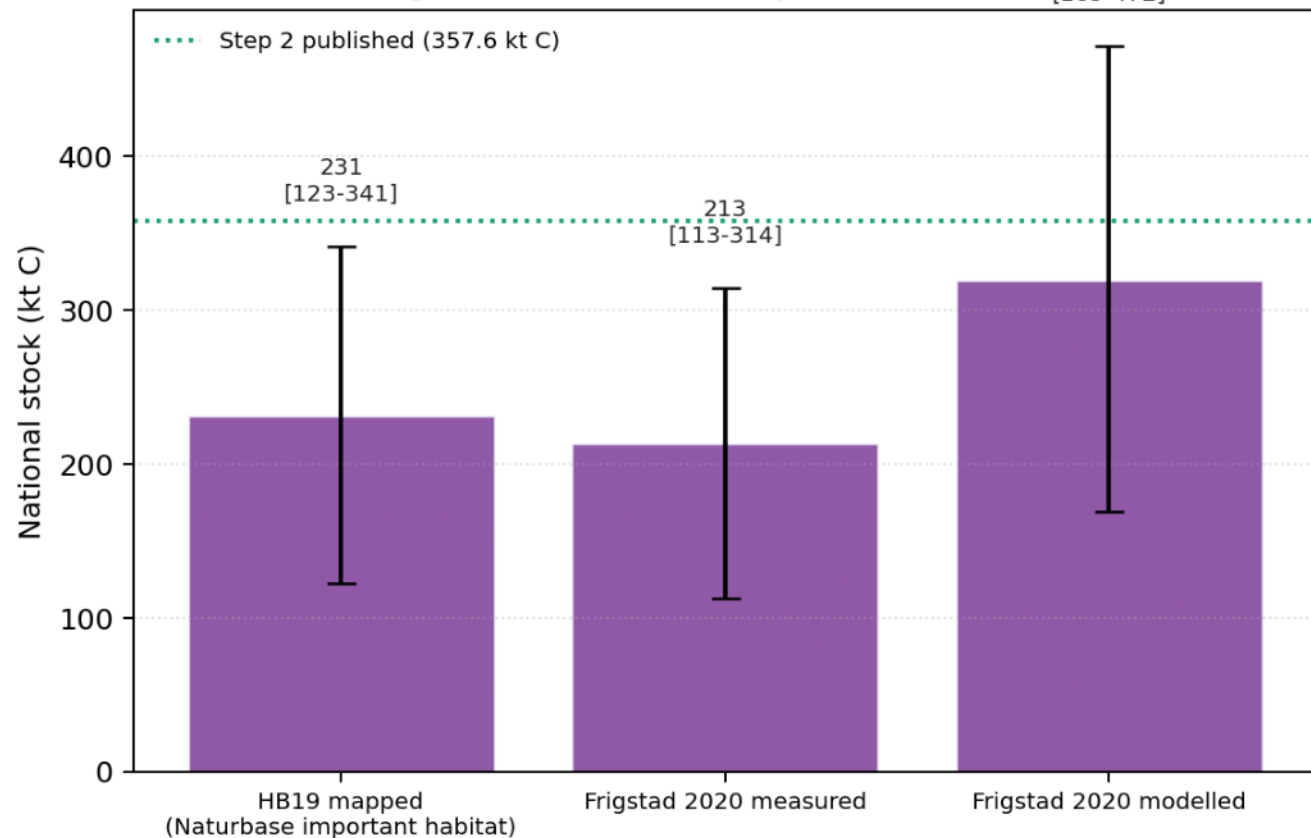


Regional disaggregation: sum = 237 kt C [151 - 316]

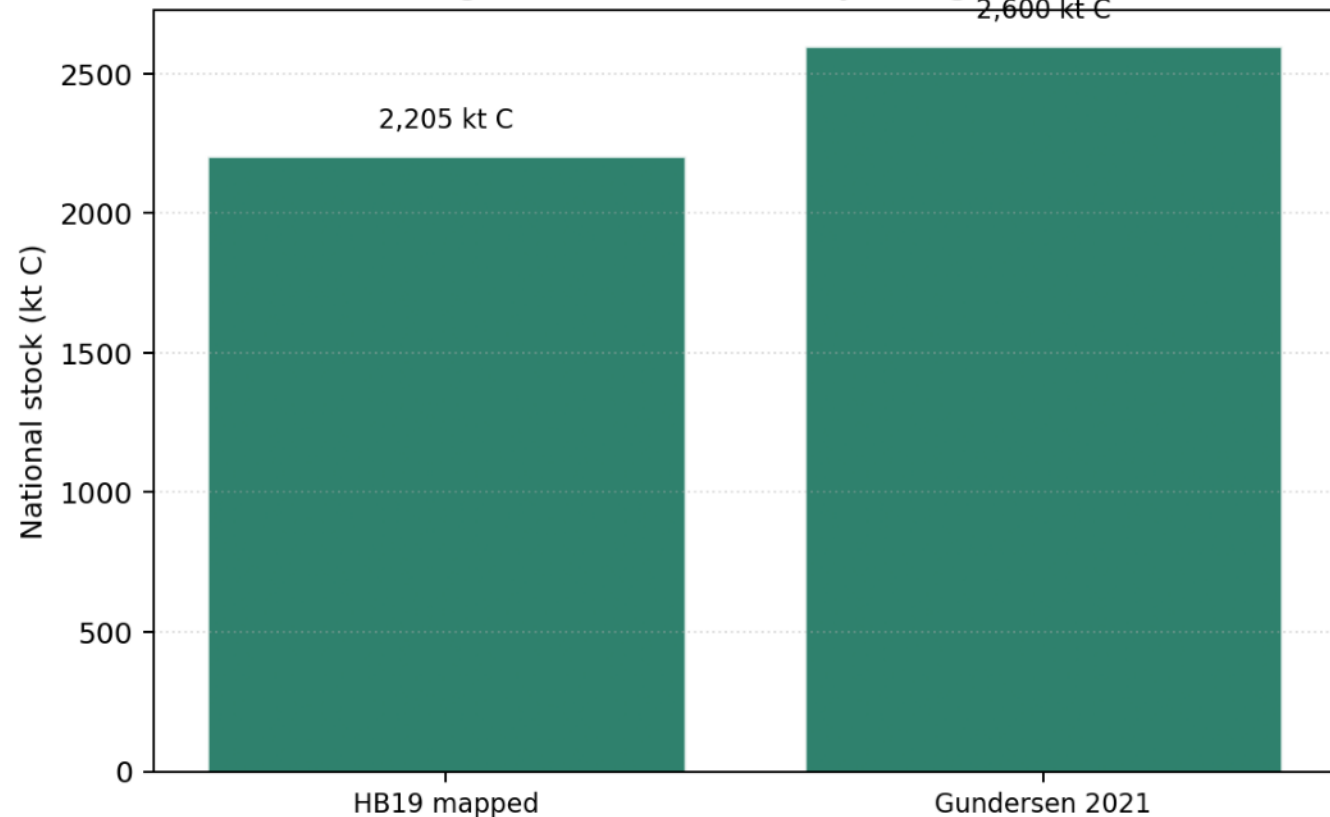


Extent sensitivity of national stock estimates

Seagrass: national stock under three extent assumptions
(stock g/m² from bootstrap of 9 measurements)



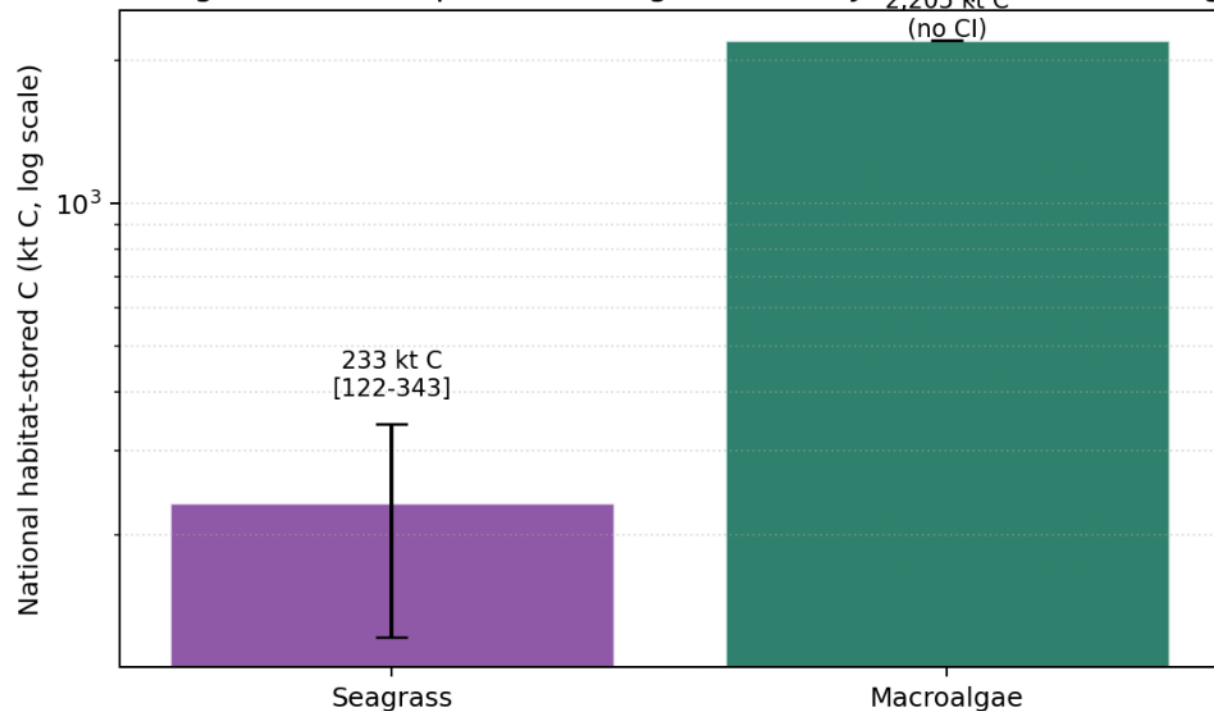
Macroalgae: national stock = Krause-Jensen density x extent
(no Norwegian field data; density 485 g C/m² fixed)



Norwegian blue-carbon balance sheet (HB19 extent; CIs where field data permit)

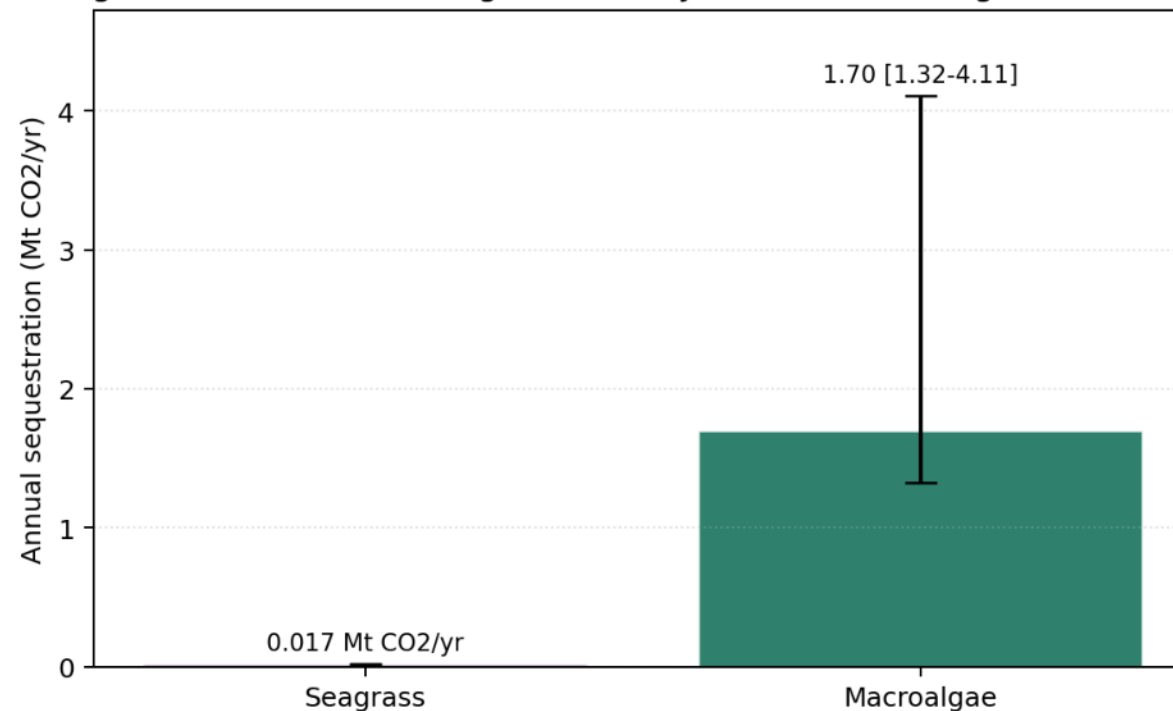
National stock (HB19 extent)

seagrass: bootstrap CI; macroalgae: Krause-Jensen point estimate (no CI)



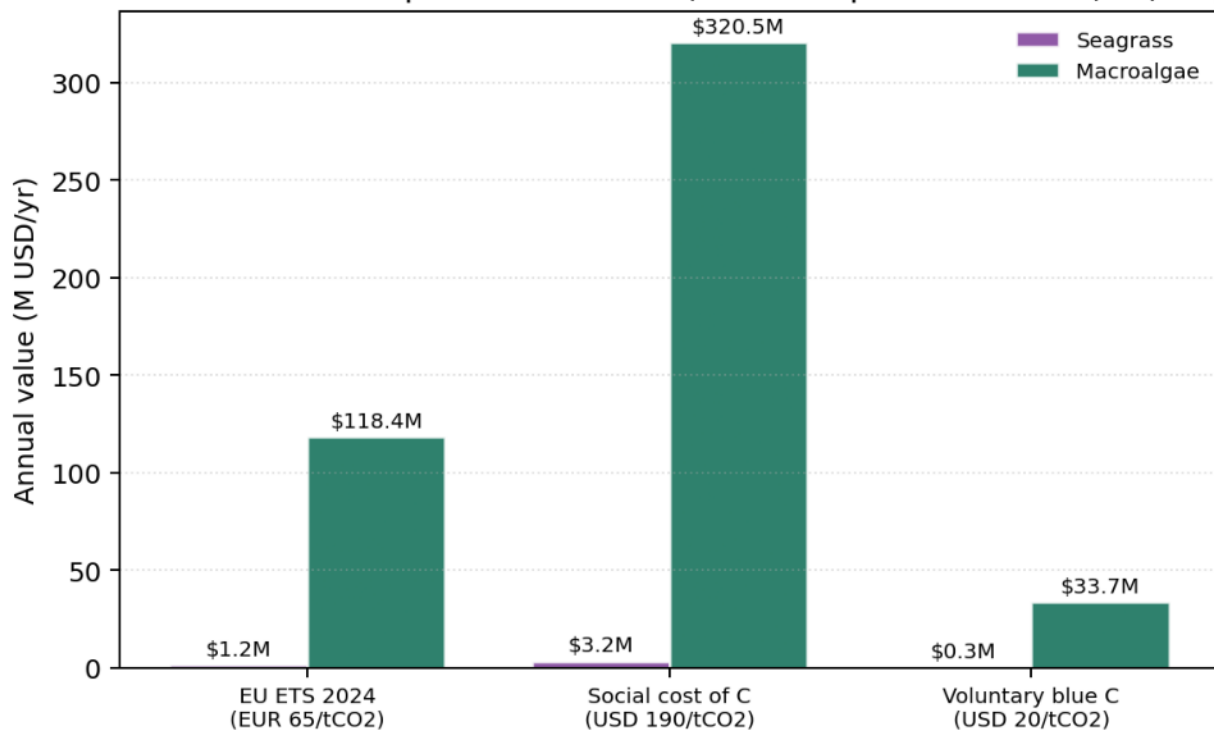
Annual sequestration

seagrass: Frigstad et al. SD; macroalgae: Krause-Jensen 25.5%, range 23-62% export fraction)

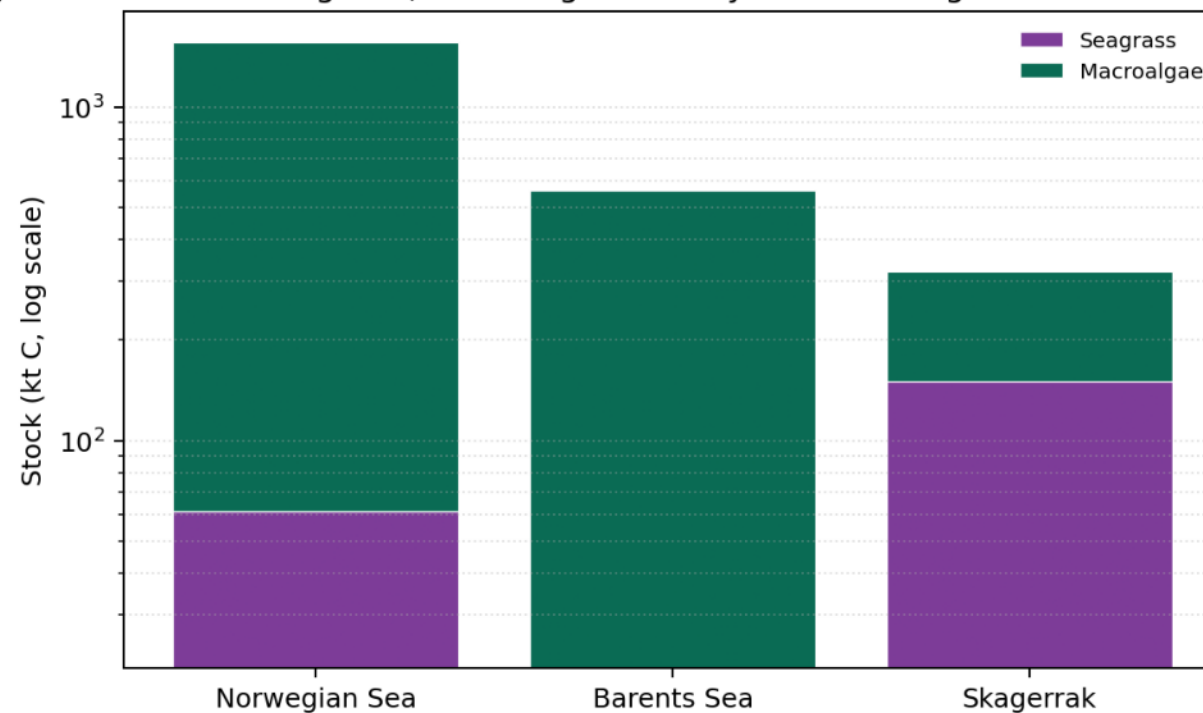


Annual sequestration value (3 carbon-price scenarios)

(seagrass stock area-weighted; macroalgae density uniform - regional variation not validated)



Regional disaggregation



Habitat-stored carbon in pressure x no-strict-protection class, by region
Seagrass error bars = bootstrap CI on stock density; macroalgae uses national mean only

